

A ALGEBRA & EQUATIONS

Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$\Delta = b^2 - 4ac$
$\Delta > 0$	2 distinct real roots
$\Delta = 0$	1 repeated real root
$\Delta < 0$	no real roots
Exponent laws	$a^m \cdot a^n = a^{m+n}$ $\frac{a^m}{a^n} = a^{m-n}$ $(a^m)^n = a^{mn}$ $a^{-n} = \frac{1}{a^n} \quad \sqrt[n]{a} = a^{1/n}$
Log laws	$\log(xy) = \log x + \log y$ $\log\left(\frac{x}{y}\right) = \log x - \log y$ $\log(x^n) = n \log x$
Change of base	$\log_a x = \frac{\log x}{\log a}$

R FINANCE, GROWTH & DECAY

Simple interest	$A = P(1 + in)$
Simple decay	$A = P(1 - in)$
Compound growth	$A = P(1 + i)^n$
Compound decay	$A = P(1 - i)^n$
Effective rate	$(1 + i_{\text{eff}}) = \left(1 + \frac{i_{\text{nom}}}{m}\right)^m$
Future value	$F = \frac{x[(1 + i)^n - 1]}{i}$
Present value	$P = \frac{x[1 - (1 + i)^{-n}]}{i}$
P = principal A = accrued amount i = rate per period n = periods x = payment m = compoundings/year	

Σ SEQUENCES & SERIES

Arith. term	$T_n = a + (n - 1)d$
Arith. sum	$S_n = \frac{n}{2}[2a + (n - 1)d]$ $S_n = \frac{n}{2}(a + l) \quad [l = \text{last term}]$
Geom. term	$T_n = a \cdot r^{n-1}$
Geom. sum	$S_n = \frac{a(r^n - 1)}{r - 1} \quad r \neq 1$
Infinite sum	$S_\infty = \frac{a}{1 - r} \quad r < 1$
a = first term d = common difference r = common ratio n = number of terms	

J DIFFERENTIAL CALCULUS

First principles	$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$
Power rule	$\frac{d}{dx}[x^n] = nx^{n-1}$
Constant	$\frac{d}{dx}[k] = 0$
Scalar multiple	$\frac{d}{dx}[k f(x)] = k f'(x)$
Sum / difference	$\frac{d}{dx}[f \pm g] = f'(x) \pm g'(x)$
$f'(x) = 0$	stationary point (max or min)
$f''(x) = 0$	point of inflection
$f'(x) > 0$	function increasing
$f'(x) < 0$	function decreasing
$f''(x) > 0$	concave up (local min)
$f''(x) < 0$	concave down (local max)

○
 TRIGONOMETRY

SOH-CAH-TOA	$\sin \theta = \frac{\text{opp}}{\text{hyp}}$	$\cos \theta = \frac{\text{adj}}{\text{hyp}}$	$\tan \theta = \frac{\text{opp}}{\text{adj}}$	$\sin(\alpha \pm \beta)$	$\sin \alpha \cos \beta \pm \cos \alpha \sin \beta$
Pythag. identity	$\sin^2 \theta + \cos^2 \theta = 1$			$\cos(\alpha \pm \beta)$	$\cos \alpha \cos \beta \mp \sin \alpha \sin \beta$
Quotient identity	$\tan \theta = \frac{\sin \theta}{\cos \theta}$			$\sin 2\alpha$	$2 \sin \alpha \cos \alpha$
Co-functions (90°)	$\sin(90^\circ - \theta) = \cos \theta$	$\cos(90^\circ - \theta) = \sin \theta$		$\cos 2\alpha$	$\cos^2 \alpha - \sin^2 \alpha$
Reduction (180°)	$\sin(180^\circ - \theta) = \sin \theta$	$\cos(180^\circ - \theta) = -\cos \theta$			$1 - 2 \sin^2 \alpha$
	$\sin(180^\circ + \theta) = -\sin \theta$	$\cos(180^\circ + \theta) = -\cos \theta$			$2 \cos^2 \alpha - 1$
Reduction (360°)	$\sin(360^\circ - \theta) = -\sin \theta$	$\cos(360^\circ - \theta) = \cos \theta$		Sine rule	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Negative angles	$\sin(-\theta) = -\sin \theta$	$\cos(-\theta) = \cos \theta$		Cosine rule	$a^2 = b^2 + c^2 - 2bc \cos A$
				Area rule	$\text{Area} = \frac{1}{2}ab \sin C$

	0°	30°	45°	60°	90°
sin θ	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan θ	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef

CAST rule — **A**ll positive (Q1) · **S**in positive (Q2) · **T**an positive (Q3) · **C**os positive (Q4) | General solution adds $n \cdot 360^\circ$ (or $n \cdot 180^\circ$ for tan), $n \in \mathbb{Z}$

○
 ANALYTICAL GEOMETRY

Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$
Parallel	$m_1 = m_2$
Perpendicular	$m_1 \times m_2 = -1$
Line (slope-int.)	$y = mx + c$
Line (point-slope)	$y - y_1 = m(x - x_1)$
Circle at origin	$x^2 + y^2 = r^2$
Circle at (a, b)	$(x - a)^2 + (y - b)^2 = r^2$

P
 PROBABILITY & COUNTING

Complement	$P(A') = 1 - P(A)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Mutually excl.	$P(A \cap B) = 0$
Independent	$P(A \cap B) = P(A) \cdot P(B)$
Factorial	$n! = n \times (n - 1) \times \cdots \times 2 \times 1$
Permutations	${}_nP_r = \frac{n!}{(n - r)!}$
Combinations	${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n - r)!}$
Counting princ.	$n_1 \times n_2 \times \cdots \times n_k$

Σ
 STATISTICS

Mean	$\bar{x} = \frac{\sum x_i}{n}$
Variance	$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$
Std deviation	$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$
Regression line	$\hat{y} = a + bx$
Slope b	$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$
Intercept a	$a = \bar{y} - b\bar{x}$
Correlation: $r \approx +1$ strong positive · $r \approx -1$ strong negative · $r \approx 0$ no correlation	

Chord & centre	Perpendicular from centre bisects chord (and conversely)	Same chord	Angles in the same segment (subtended by the same chord) are equal
Angle at centre	Angle at centre = $2\times$ angle at circumference (same arc)	Tangent $\perp$ radius	Tangent is perpendicular to radius at point of tangency
Semicircle	Angle in a semicircle = 90°	Tangent–chord	Tangent–chord angle = inscribed angle in alternate segment
Cyclic quad.	Opposite angles of a cyclic quad. are supplementary (sum = 180°)	Equal tangents	Two tangents from an external point are equal in length

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